

Date: 02-02-2026

```
##string
##str="20"
##print(str)
##q= int(str)
##print(q)
```

```
##c=int(input("c="))
##print(c)
```

```
##concat
##a ="Monkey"
##b= "D luffy"
##print(a+b)
```

```
#-----
-----
```

```
## operators
##Arithmetic operator
##Assaignment operator
##comparison operator
##logical operator
##bitwise operator
##identify operator
##membership operator
```

```
#-----
-----
```

```
##Arithmetic operator
```

```
## + - * / %
```

```
'''
```

```
a=4
b=6
print(a+b)
print(a-b)
print(a*b)
print(a**b)
print(a//b)
print(b//a)
print(b%a)
print(a%b)
'''
```

```
#-----
-----
```

```
##Assaignment operator
```

```
## = != += -= *= /= //
```

```
##a=6
###a+=5
###a**=2    can only used here not inside print
## a/b
```

```
###a//=2    # // to not print op in float
```

```
##
```

```
##print(a)
```

```
#-----
```

```
## comparison operator
```

```
date:03.02.2026
```

```
## return boolean values
```

```
## == <= >= != < >
```

```
##a= 10
```

```
##b= 15
```

```
##print(10==10)
```

```
##print(10==11)
```

```
##
```

```
##print(a<=b)
```

```
##print(50<=10)
```

```
##
```

```
##print(10>=5)
```

```
##print(5>=10)
```

```
##
```

```
##print(10!=20)
```

```
##print(10!=10)
```

```
##
```

```
##print(4<5)
```

```
##print(5)
```

```
#-----
```

```
## logical operator
```

```
## (Gate) and or not      (&& || !)
```

```
##a=10
```

```
##b=20
```

```
#and (can't put capital)
```

```
#3print(a>b & b<a)
```

```
##print(a>b and b>a)
```

```
##print(a<b and b<a)
```

```
##print(a<b and b>a)
```

```
## #or
```

```
##print(a>b or b<a)
```

```
##print(a>b or b>a)
```

```
##print(a<b or b!=a)
```

```
##print(a<b or b>a)
```

```
##
```

```
## #not
##print(a==b)
##print(not(a==b))
##print (a!=b)
##print(not(a!=b))
```

```
#-----'
```

```
#bitwise operator
#& and gate
```

```
##a=10
##b=12
##print(a&b)
##print(bin(8))
## | or gate
##a=3
##b=4
##print(a|b)
##print(7)
```

```
## ^ xor gate
```

```
##a=2
##b=3
##print(a^b)
```

```
##p=4
##q=5
##print(a^b)
```

```
## ~ bitwise not gate  $\sim x = -(x+1)$ 
```

```
##f=5
##print(~(f)) # $\sim(-5+1) = -6$ 
##print(bin(~(f)))
##print(bin(-6))
##
##print(bin(~f & 0xff)) # to get cpu's 8-bit values
```

```
#-----
```

```
##membership operator          Date:04-02-2026
```

```
## in
## not in
'''
```

```
a = "python"
print("o" in a)
print("a" in a)
```

```
print("o" not in a)
print("a" not in a)
'''
```

```
#-----
```

identity operator

##is (check the memory address) == ()
##is not

##z="zoro"
##print(id(z))
"

a=10
b=11
c=a

print(a is b)
print(a is c)

print(id(a))
print(id(b))
print(id(c))

print(a == b)
print(a == c)

"

is not
"

a="luffy"
b="zoro"
c="luffy"

print(a is not b)
print(a is not c)
print(id(a))
print(id(b))
print(id(c))
"

#-----

#-----

string methods

#upper
"

a="abc"
print(a.upper())

a="Abc".upper()
print(a)

print("Zoro".upper())

```
b=input("Enter your name: ").upper()
print(b)
```

```
"""
# lower
```

```
a="Python"
print(a.lower())
```

```
a="DEC".lower()
print(a)
```

```
c=input("Enter your name: ")
print(c)
```

```
"""
# swap
```

```
##a="sanji".swapcase()
##b="SANJI".swapcase()
##print(a,b)
```

```
# capitalize
```

```
##a="i am watching,
##i am eating".capitalize()
##print(a)
```

```
#title
##a="the one piece.
##the one piece is a fiction story".title()
##print(a)
```

```
"""
# length()
```

```
a="abcd"
print(len(a))
```

```
a=input("Enter your name: ")
print(len(a))
```

```
#split()
```

```
a="Monkey D Luffy"
print(a.split("o"),a.split("f"))
```

```
#replace()
```

```
a="Black leg Sanji"
print(a.replace("Black leg","Vinsmoke"))
```

```

'''
#-----

#Ascii values
'''

A=65
Z=90

a=97
z=122

'''      # max, min, ord, chr

##print(max("Anbu"))  # To find max num values in Ascii
##print(min("Anbu"))  # To find min values in Ascii
##print(ord('z'))     # To find ascii num value of a variable
##print(chr(97))      # To find variable of a ascii num value
#-----

# .isalpha(), .isdigit(), .isupper() & islower()      Date:05-02-2026

a="python"
b="18"
c="SANJI"
'''

print(a.isalpha())
print(b.isalpha())

print(a.isdigit())
print(b.isdigit())

print(c.isupper())
print(a.isupper())

print(a.islower())
print(c.islower())
'''
'''
# .join()
print("0".join(b))
# .strip
d= "      leo      "

print(d)
print(d.strip())
# .count()
print(a.count("y"))
'''

# endwith(), startswith(), sep
'''

```

```

e= "one piece is peak"
print(e.endswith("peak")) # to check True or False end with the word
print(e.endswith("one"))

print(e.startswith("one")) # to check True or False start with the word
print(e.startswith("peak"))

print("hi", "hello")
print("hi", "hello",sep="_")

print(c)
print(c, end="!")    #joins in end without space
"""

```

#Positive index

```

f="python"

"""      # 0 1 2 3 4 5
print(f[0])      # p y t h o n
print(f[1])      # -6 -5 -4 -3 -2 -1
print(f[5])
"""

##print(f[6]) # error : out of range

"""

print(f[0:5]) #start & end

print(f[:3]) #end # end value not print in op : pytho
print(f[:6])

print(f[:]) #

print(f[3:]) #start
print(f[5:])

"""

# step
"""
print(f[0:6:1]) #straight #last ratio is times to read
print(f[6:0:-1]) #reverse
print(f[0:5:2]) # x-ple
print(f[::2])
"""

```

```

#Negative index          # p y t h o n
#-6 -5 -4 -3 -2 -1
'''
print(f[-0])      # 0 or -0 == p
print(f[-1])
print(f[-6])
'''

#print(f[-7])      # error : out of range
'''
print(f[-6 : -1])  #start and end
print(f[-0 : -1])
print(f[-6 : -3])

print(f[ : -1])    #end # end value not print in op : pytho
print(f[ : -6])

print(f[ : ])      #

print(f[-6 : ])    #start
print(f[-1 : ])

'''
'''
# step

print(f[-6:-1:1])  #straight
print(f[-1:-6:-1]) #revrse
'''

```

Conditional statement

if, else, elif

```

'''
if True:    # True == 1
print("true")
else:
print("false")

if False:   # False == 0
print("true")
else:
print("false")

'''
'''
if '':      # " " == true and "" == false without space false
print("True")

```

```
else:  
print("False")  
"""
```

```
#pass oor fail  
"""
```

```
a=int(input("ENTER YOUR MARK: "))
```

```
if a>34:  
print("Great you are pass")  
else:  
print("Sorry you are fail")  
"""
```

```
# To check odd or even  
"""
```

date:06-02-2026

```
a=int(input("Enter a Number: "))
```

```
if a%2 == 1:  
print("odd")  
else:  
print("even")  
"""
```

```
# to check palindrome:  
"""
```

```
a=input("enter a word: ")  
print(a[::-1])
```

```
if a == a[::-1]:  
print("palindrome")
```

```
else:  
print("Is not palindrome")  
"""
```

```
# To check Leap year: #####  
"""
```

```
a=int(input("Enter a year: "))
```

```
if a%4 == 0 and a%100 != 0 or a%400 == 0:  
print("Leap year")
```

```
else:  
print("It is not Leap year")  
"""
```

```
# To check vowels in or not  
"""
```

```
a=input("Enter A word: ").lower()
```

```
if "a" in a or "e" in a or "i" in a or "o" in a or "u" in a:
    print("vowels")
else:
    print("constants")
"
```

```
#
```

```
# Nested if:
```

```
"
m = int(input("Enter a Number: "))
```

```
if m > -1 and m < 101:
```

```
if m > 34:
    print("pass")
else:
    print("fail")
```

```
else:
    print("Invalid Input")
```

```
"
```

```
"
```

```
a=int(input("Enter a year: "))
```

```
if a>-1 :
if a%4 == 0 and a%100 != 0 or a%400 == 0:
    print("Leap year")
else:
    print("Is not a Leap Year")
else:
    print("invalid input")
"
```

```
"
```

```
i = int(input("Enter a Number: "))
```

```
if i%3 == 0 :
if i%5 == 0:
    print(i,"is divisible by both 3 and 5")
```

```
else:
    print(i," is divisible by 3 only")
else:
    print(i," is not divisible by 3")
"
```

```
# grade terms
```

```
## 90 - 100 A
## 80 - 89 B
```

```
## 70 - 79 C
## 60 - 69 D
## 50 - 59 E
## 0 - 49 RA
```

```
# elif:
'''
```

```
j = int(input("Enter your Mark: "))
```

```
if 90 <= j <= 100:
print("GRADE A")
elif 80 <= j <= 89:
```

```
print("GRADE B")
elif 70 <= j <= 79:
print("GRADE C")
elif 60 <= j <= 69:
print("GRADE D")
elif 50 <= j <= 59:
print("GRADE E")
elif 0 <= j <= 49:
print("RA")
else:
print("Invaalid Mark")
'''
```

```
#-----
```

```
# To find max value: # Date:07-02-2026
```

```
##a=int(input("Enter a number for a: "))
##b=int(input("Enter a number for b: "))
##c=int(input("Enter a number for c: "))
'''
```

```
if a<b:
print("b is greater than a")
elif a>b:
print("a is greater than b")
else:
print("a == b")
'''
'''
```

```
if a>b and a>c:
print(a," is the highest value")
elif b>a and b>c:
print(b, "is the highest value")
elif c>a and c>b:
print(c, "is the highest value")
else:
print("a,b and c are same")
'''
```

```
##un="sanji"
##pas="abc123"
'''
```

```
a=(input("Enter a un for a: "))
b=(input("Enter a pas for b: "))
```

```
if a == un and b==pas:
print(" login successfull")
elif b != pas and un==a:
print("password is incorrect")
elif un!=a and b==pas:
print("username unmatched")
else:
print("Both are wrong")
'''
```

#Assesment:

```
# 1. +,-,*,/,//,**,%
# 2. leap year
# 3. palindrome
# 4. odd or even
# 5. find the highest value
# 6. grade
```

```
##a=int(input("Enater a num a: "))
##b=int(input("Enter a num b: "))
##c=(input("Enter a operator: "))
```

```
## 1. operator +,-,*,/,//,**,%:
```

```
'''
```

```
if c == "+":
print("sum of a + b =: ",a + b )
elif c == "-":
print("sutract of a - b: ",a - b)
elif c == "*":
print("Multple of a*b: ",a*b)
elif c == "**":
print("Power a**b: ",a**b)
elif c == "/" and b==0:
print("undifined")
elif c == "/":
print("Division of a/b: ",a/b)
elif c == "//" and b==0:
s print("undefined")
elif c == "//":
print("Division of a//b: ",a//b)
elif c == "%" and b==0:
print("undefined")
elif c == "%":
print("Reminder of a%b: ",a%b)
else:
print("Invalid operator")
'''
```

```
## 2. leap year
```

```
'''
```

```
year = int(input("Enter a year: "))
if year%4 == 0 or year%100 != 0 and year%400 == 0:
```

```
print("Leap year")
else:
print("Not Leap year")
'''
```

```
## 3. palindrome
'''
```

```
pal = input("Enter a word: ")
rev = pal[::-1]
if pal == rev:
print("plindrome")
else:
print("Not Palindrome")
'''
```

```
## 4. odd or even
'''
```

```
oe = int(input("Enter a Num: "))
```

```
if oe%2 == 1:
print("odd")
else:
print("even")
'''
```

```
## 5. find the maximum number
'''
```

```
x = int(input("Enter a num x: "))
y = int(input("Enter a num y: "))
z = int(input("Enter a num z: "))
```

```
if z<x>y:
print("x is max")
elif x<y>z:
print("y is max")
elif x<z>y:
print("z is max")
else:
print("all are same")
'''
```

```
## 6. Grade
''''''
```

```
Mark = int(input("Enter your Mark: "))
```

```
if 89 < Mark <101: #90-100
print("Grade A")
elif 79 < Mark < 90: #80-89
print("Grade B")
elif 69 < Mark < 80: #70-79
print("Grade C")
elif 59 < Mark < 70: #60-69
print("Grade D")
elif 49 < Mark < 60: #50-59
print("Grade E")
elif 34 < Mark < 50: #35-49
```

```
print("pass")
elif -1 < Mark < 35: #0-34
print("Fail")
else:
print("Invalid input")
"""
```

```
#
```

```
# Holiday on sunday Date:08-02-2026
```

```
#
```

```
# single line if Date:09-02-2026
```

```
# odd or even
##a=int(input("enter a: "))
##if a%2==0:
## print("even")
##else:
## print("odd")
```

```
# sinle line odd or even
##q=int(input("enter q: "))
##q = "even" if q%2==0 else "odd"
##print(q)
```

```
#divisible by 3 and 5 single line
```

```
##d=int(input("Enter d: "))
##d = ("divisible by 3 and 5" if d%5==0 else "divisible by 3 only") if d%3==0 else " not by 3"
##print(d)
```

```
#
```

```
#3 for loop
"""
```

```
for i in range(5): #end
print(i)
"""
```

```
##for i in range(3,5): #stadn, ert
## print(i)
##
```

```
##for i in range(0,10,2): ## (start,end("but last vlue not print like index"), step)
## print(i)
```

```
##for i in range(10,0,-1): #for reverse print also like index
```

```

## print(i)

# even number
"""
n=int(input("Enter n: "))
for i in range(0,n,2):
    print(i)
"""

#string

##for i in "sanji":
##    print(i)

##
##a="luffy"
##for j in a:
##    print(j)

##st = input("Enter aa string: ")
##for k in st:
##    print(k)

##z="python"
##rev = z[::-1]
##
##for l in rev:
##    print(l)

# string reverse iterate
"""
a="python"
for i in range(len(a)-1,-1,-1): # here do hte reverse
    print(a[i])
"""

# another method
"""
a="python"
for i in range (1,(len(a)+1),1):
    print(a[-i]) # here do the reverse
"""

# Tables

##t=(int(input("Enter a num: ")))
##for i in range(1,11):
##    print(i, "*", t,"=",i*t)

# print even numbers only
"""
e=(int(input("Enter a num: ")))
for i in range(e):
    if i%2 == 0: # if i%2==1 , it will print odd nums only

```



```

print(i)
'''

#sum of numbers:
#if n=5 op == 15    0+0 = 0    #back process
#0+1= 1
#1+2= 3
# 3+3= 6
#6+4= 10
# 10+5=15
'''

```

```

n=int(input("enter num: "))
count=0
for i in range(1,n+1):
count = count+i
print(count)
'''

```

```

# counting vowels
'''
a=input("Enter a word: ").lower()
count = 0
for i in a:
#if "a" in i or "e" in i or "i" in i or "o" in i or "u" in i:
# or
if i in "aeiou":
count= count+1    # count+=1
print(count)
'''

```

#

Assesment: only explanation

Date:10-02-2026

###

#1. factorial # self made

```

'''
f=int(input("Enter a num: "))
fac = 1
for i in range(f,0,-1):
fac=i*fac
print("Factorial of",f, ":", fac)
'''

```

#2. perfect number {ex: 6,8,28,128,426

#1 + 2 + 3 = 6. and 1,2,3 are divisor of 6 # so 6 is perfect number

#1 + 2 + 4 + 7 + 14 = 28. 1,2,4,7,14 are divisor of 28 # so 28 is the perfect number

'''

```

p=int(input("Enter a num: "))
count=0
for i in range(1,p):
    if p%i == 0:
        count+=i

```

```

if count == p:
    print("perfect number")
else:
    print("Not perfect number")
'''

```

#3. Armstrong Number:

153 (3 digits): Calculate $1^3 + 5^3 + 3^3$. $1 + 125 + 27 = 153$. Since it matches, 153 is an Armstrong number.

9474 (4 digits): Calculate $9^4 + 4^4 + 7^4 + 4^4$. $6561 + 256 + 2401 + 256 = 9474$. Since it matches, 9474 is an Armstrong number.

10 (2 digits): Calculate $1^2 + 0^2$. $1 + 0 = 1$. Since $1 \neq 10$, 10 is not an Armstrong number.

Key Facts:

1. 0-9 are armstrong numbers
2. there is no two digits are armstrong numbers

```

n=int(input("Enter a num: "))
s=str(n)
digit=len(s)
count=0
for i in s:
    count= count+int(i)**digit
if count == n:
    print("Armstrong Numer")
else:
    print("Not Armstrong number")
'''

```

#4. sum of digits:

```

a=int(input("Enter a num"))
s=str(a)
count=0

```

```

for i in s:
    count=count+int(i)
print("sum of digits: ",count)
'''

```

#5. fibonacci series

$F_n = F_{n-1} + F_{n-2}$

The Fibonacci series is a sequence of numbers where each number is the sum of the two preceding ones. It usually starts with 0 and 1.

1. The Sequence

The series progresses like this: 0, 1, 1, 2, 3, 5, 8, 13, 21, 34, 55, 89, 144...

"""

a,b = b, a+b

"""

f=int(input("Enter a num: "))

a=0

b=1

for i in range(1,f+1):

print(a)

a,b=b,a+b

"""

reverse string:

"""

n="python"

a=""

for i in n:

a=i+a

print(a)

"""

assessmennt:

#fascinating number: (192,219,273,327)

"""

192

Step 1: 192 \times 2 = 384

Step 2: 192 \times 3 = 576

Step 3: Concatenate them: 192384576

Result: This sequence contains 1, 2, 3, 4, 5, 6, 7, 8, and 9 exactly once.

Therefore, 192 is a fascinating number.

""" # completed

code in 2__for and while loop__statements.py file

#

#

Date:11-02-2026

